INCH-POUND

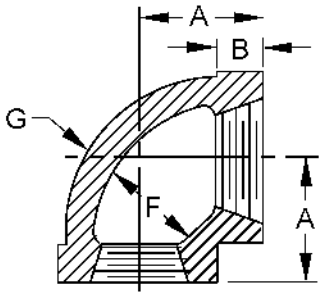
MS14311D
w/AMENDMENT 1
10 July 2025
SUPERSEDING
MS14311D
1 November 2015

DETAIL SPECIFICATION SHEET

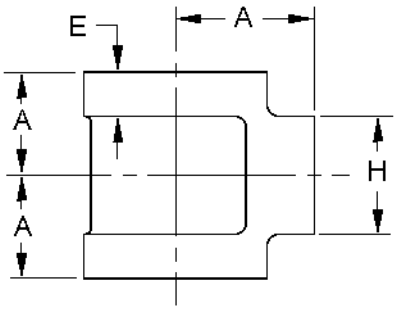
FITTINGS, PIPE, BRASS OR BRONZE, THREADED (90-DEGREE ELBOWS, TEES, CROSSES, 45-DEGREE ELBOWS, AND COUPLINGS), CLASS 250

This specification is approved for use by all Departments and Agencies of the Department of Defense.

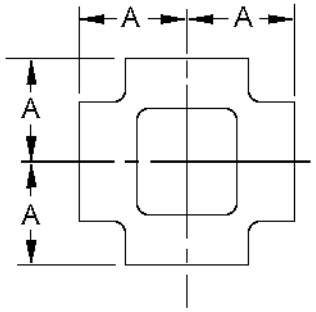
The requirements for acquiring the product described herein shall consist of this specification sheet and WW-P-460.



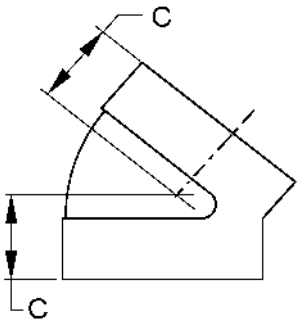
P - 90-DEGREE ELBOW



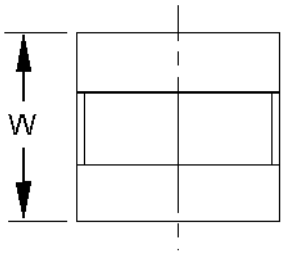
Q - TEE



R - CROSS



S - 45-DEGREE ELBOW



T- COUPLING

FIGURE 1. Elbows, tees, crosses and coupling fittings.

AMSC N/A
DISTRIBUTION STATEMENT A: Approved for public release; Distribution is unlimited.

FSC 4730

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Dash number to pipe size.

Dash number	Pipe size inch (fraction)		mm
- 1	1/4	(.250)	6.35
- 2	3/8	(.375)	9.53
- 3	1/2	(.500)	12.70
- 4	3/4	(.750)	19.05
- 5	1	(1.000)	25.40
- 6	1-1/4	(1.250)	31.75
- 7	1-1/2	(1.500)	38.10
- 8	2	(2.000)	50.80
- 9	2-1/2	(2.500)	63.50
- 10	3	(3.000)	76.20
- 11	4	(4.000)	101.60

NOTES:
1. Dimensions are in inches,
2. Metric equivalents are given for information only.
3. Material: Brass or bronze in accordance with ASME-B16.15.
4. Dimensions: In accordance with ASME-B16.15, table titled “Dimensions of 90-deg Elbows, Tees, Crosses, 45-deg Elbows, and Couplings (Straight Sizes) - Class 250”.

FIGURE 1. Elbows, tees, crosses and coupling fittings – Continued.

Requirements

Material:

Non-potable water: Brass or bronze (leaded) in accordance with ASME-B16.15.

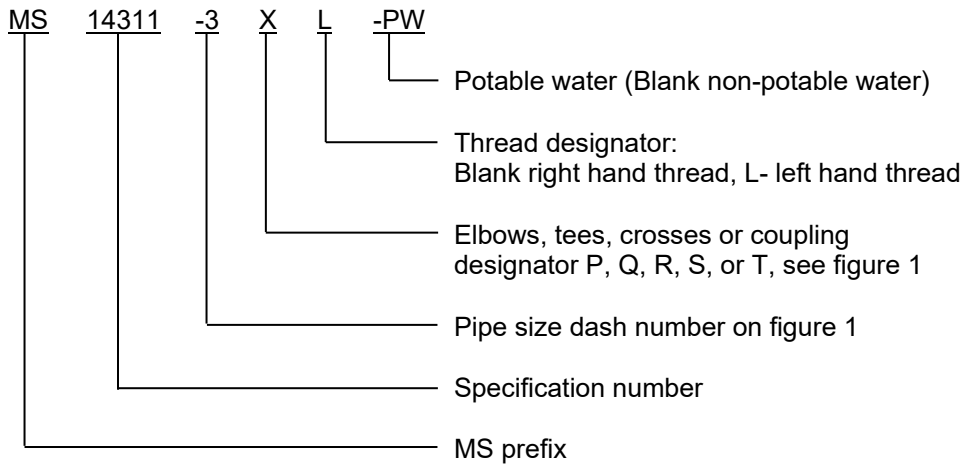
Potable water: Brass alloys C89510 or C89520 in accordance with ASTM B584 or C89550 in accordance with Copper Development Association Inc.

Threads: In accordance with ASME-B16.15.

Protective coating: None.

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PIN: The PIN consists of the MS prefix, specification number, dash number for size, 90° or 45° elbows, tee, cross or coupling designator, right hand thread designator (Blank), non-potable water (Blank) or potable water dash letters (-PW). When left-hand threads are required, add the letter “L” to the PIN.



Examples:

MS14311-5Q represents a 1 inch Nominal Pipe Size (NPS) tee, non-potable water.
MS14311-5Q-PW represents a 1 inch NPS tee, potable water.
MS14311-8SL represents a 2 inch NPS 45° elbow, left-hand threads, non-potable water.
MS14311-8SL-PW represents a 2 inch NPS 45° elbow, left-hand threads, potable water.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Brass fittings for potable water use may be substituted for non-potable water brass fittings. Non-potable brass fittings shall not be substituted for potable water fittings.

January 4, 2014, Federal Public Law 111-380 prohibits the sale of plumbing fixtures with a wetted surface area containing more than a weighted average of 0.25% lead.

Marking: In accordance with ASME-B16.15, additional marking for potable water fittings shall be as described below.

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Potable water fittings. Brass 90° elbows, tees, crosses, 45° elbows, and couplings for potable water systems shall be certified and marked as specified:

Many State plumbing codes require the use of products that are in compliance with NSF/ANSI standards 61 and 372. NSF/ANSI Standard 372 is a certification process by which independent laboratories verify that the plumbing product is in compliance with the requirements of the 2011 Reduction of Lead in Drinking Water Act. Products will bear the mark of the laboratory that has independently certified that the product meets the standard. EPA published a brochure to assist the public with identifying the various marks that indicate a product has been certified as lead free to the new requirement of the Act: "How to Identify Lead Free Certification Marks for Pipes, Fittings, Fixtures, Solder, and Flux Used for Drinking Water." You can access the document at How to Identify Lead Free Certification Marks for Pipes, Fittings, Fixtures, Solder, and Flux Used for Drinking Water | US EPA.

Non-potable water fittings. Brass 90° elbows, tees, crosses, 45° elbows, and couplings not for potable water systems shall be tagged:

NOT FOR USE IN POTABLE WATER SYSTEMS.

The tag shall be removable by the end item user.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to WW-P-460, this document references the following:

Reduction of Lead in Drinking Water Act
ASME-B16.15
ASTM B584
NSF/ANSI standards 61
NSF/ANSI Standard 372

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CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - MC
Air Force – 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2025-049)

Review activities:
Army - AR, CE, EA
Navy - SA, SH, YD

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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